

Lab 07

Due March 17, 5:00 PM PST

AUTHOR

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Problem 1a

```
## write your answer here
my_dist <- function(data, centroids){
  n_data <- nrow(data)
  k <- nrow(centroids)
  dist_mat <- matrix(NA, nrow = n_data, ncol = K)
  for(i in 1:n_data){
    for (k in 1:K){
      dist_mat[i,k] <- dist(rbind(data[i,], centroids[k, ]))
    }
  }
  dist_mat
}
```

Problem 1b

```
## write answer here
my_dist <- function(x, y) {
  sqrt(sum((x - y)^2))
}

my_kmeans <- function(data, k, max_iter = 100, threshold = 1e-6) {
  data <- as.matrix(data)

  n <- nrow(data)
  d <- ncol(data)

  init_idx <- sample(1:n, k)
  centroids <- data[init_idx, , drop = FALSE]

  clusters <- rep(NA, n)

  for (iter in 1:max_iter) {
    old_centroids <- centroids
```

```

for (i in 1:n) {
  distances <- rep(NA, k)
  for (j in 1:k) {
    distances[j] <- my_dist(data[i, ], centroids[j, ])
  }
  clusters[i] <- which.min(distances)
}

for (j in 1:k) {
  cluster_points <- data[clusters == j, , drop = FALSE]

  if (nrow(cluster_points) > 0) {
    centroids[j, ] <- colMeans(cluster_points)
  }
}

centroid_shifts <- rep(NA, k)
for (j in 1:k) {
  centroid_shifts[j] <- my_dist(old_centroids[j, ], centroids[j, ])
}

if (all(centroid_shifts < threshold)) {
  break
}

return(list(
  centroids = centroids,
  clusters = clusters
))
}

```

`my_kmeans(data, k, max_iter = 100, threshold = 1e-6)` performs k-means clustering on a dataset of arbitrary dimension, where `data` is a numeric matrix or data frame with observations in rows and variables in columns, `k` is the number of clusters, `max_iter` is the maximum number of iterations allowed before the algorithm stops, and `threshold` is the convergence cutoff such that the algorithm stops when all Euclidean distances between old and updated centroids are smaller than this value. The function returns a list with two components: `centroids`, a matrix containing the final cluster centers, and `clusters`, a vector giving the cluster assignment of each observation from the last iteration.

Problem 1c

```

## write answer here
set.seed(123)
peng_scaled <- read.csv("peng_scaled.csv")

```

```
my_fit$clusters
```

```
base_fit$cluster
```

[illegible]